

Three forms of information transfer in neural architectures

Comparative methods section

Connections that bypass or join computational streams serve different purposes in the original Transformer, ResNet, and U-Net. Their interpretation depends on the operation at the receiving node: attention uses queries to access a memory, residual addition combines corresponding tensor elements, and concatenation retains features as separate channels. Figures 1–3 expose these distinctions at the module level.

Sequence conditioning. The original Transformer encoder–decoder [1] adds positional encodings to source and shifted-right target embeddings. Its six encoder layers each apply unmasked multi-head self-attention followed by a position-wise feed-forward network. Each of six decoder layers instead applies masked self-attention, encoder–decoder attention, and a feed-forward network. Every sublayer has a residual addition followed by layer normalization. Masking excludes future target positions. In cross-attention, queries come from the preceding decoder sublayer, while keys and values come from the final encoder output; this same memory supplies every decoder layer (Fig. 1). Thus, source information conditions target processing through attention, separately from the local residual paths. The base model uses width 512, eight attention heads, and feed-forward width 2048. A learned linear projection and softmax turn the final decoder output into next-token probabilities.

Residual refinement. For the original same-shape ResNet blocks [2], the input x branches into a learned residual $F(x)$ and an identity shortcut, giving $y = \text{ReLU}(x + F(x))$. Elementwise addition requires compatible shapes and does not increase the channel count. Figure 2 compares the basic block’s two 3×3 convolutions at 64 channels with the bottleneck’s 1×1 , 3×3 , and 1×1 convolutions, taking channels from 256 to 64 to 64 to 256. Batch normalization follows every convolution; no activation separates the final normalization from addition. ReLU follows the merge. These examples preserve spatial resolution: dimension-changing blocks require shape matching and are outside this comparison. Basic and bottleneck blocks underlie ResNet-18/34 and ResNet-50/101/152, respectively.

Spatial feature fusion. Original U-Net [3] uses four contracting stages and a bottom stage, with paired valid 3×3 convolutions and ReLU. Four stride-two pooling operations reduce resolution. Each expanding stage uses a 2×2 up-convolution, halving channels, then concatenates a cropped encoder copy before two further convolutions. Cropping aligns spatial sizes reduced by valid convolutions; concatenation doubles the upsampled channel count (Fig. 3). The four copied side lengths change from (568, 280, 136, 64) to (392, 200, 104, 56). A 572×572 input produces a 388×388 output, with a final 1×1 convolution mapping 64 channels to two classes.

The cross-stream operations therefore impose different interfaces. Transformer memory contributes keys and values to a query-dependent operation; ResNet directly sums equal-shaped representations; U-Net aligns spatial extents while appending channels. Calling all three connections “skip links” would obscure both their computational roles and the tensor constraints needed to reproduce them.

References

- [1] Vaswani et al. *Attention Is All You Need*. NIPS, 2017. Fig. 1; §§3.1–3.5.
- [2] He et al. *Deep Residual Learning for Image Recognition*. CVPR, 2016. Figs. 2, 5; §§3.2–3.4, 4.1.
- [3] Ronneberger, Fischer, and Brox. *U-Net: Convolutional Networks for Biomedical Image Segmentation*. MICCAI, 2015. Fig. 1; §2; arXiv v1.

Transformer

2017 base model · width 512 · 8 heads

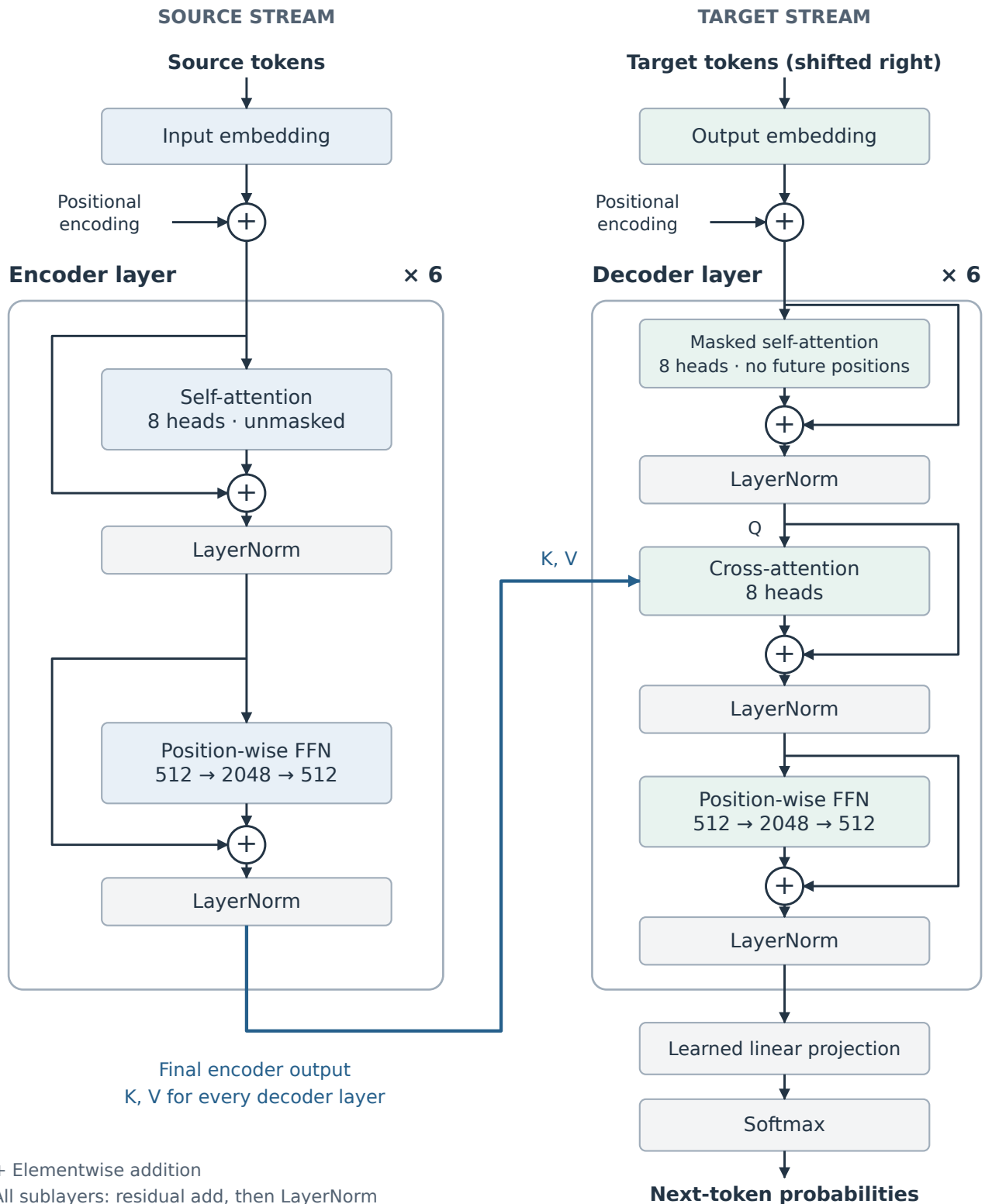
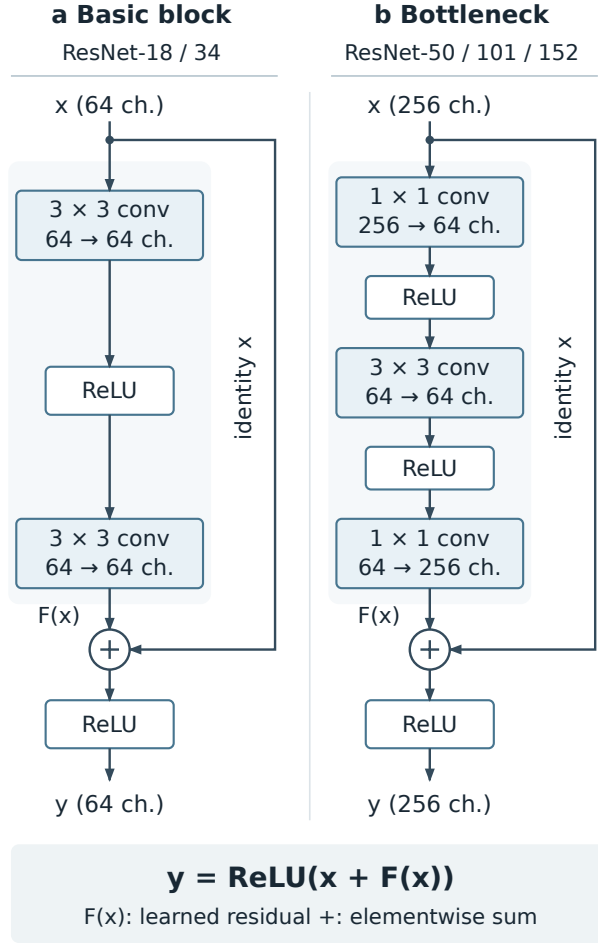


Figure 1: Original Transformer, base model. Each container represents six successive layers. The final encoder memory supplies keys and values to cross-attention in every decoder layer; queries arrive from the preceding decoder sublayer. Local bypasses add before LayerNorm. FFN denotes a position-wise feed-forward network. Based on Vaswani et al. [1].



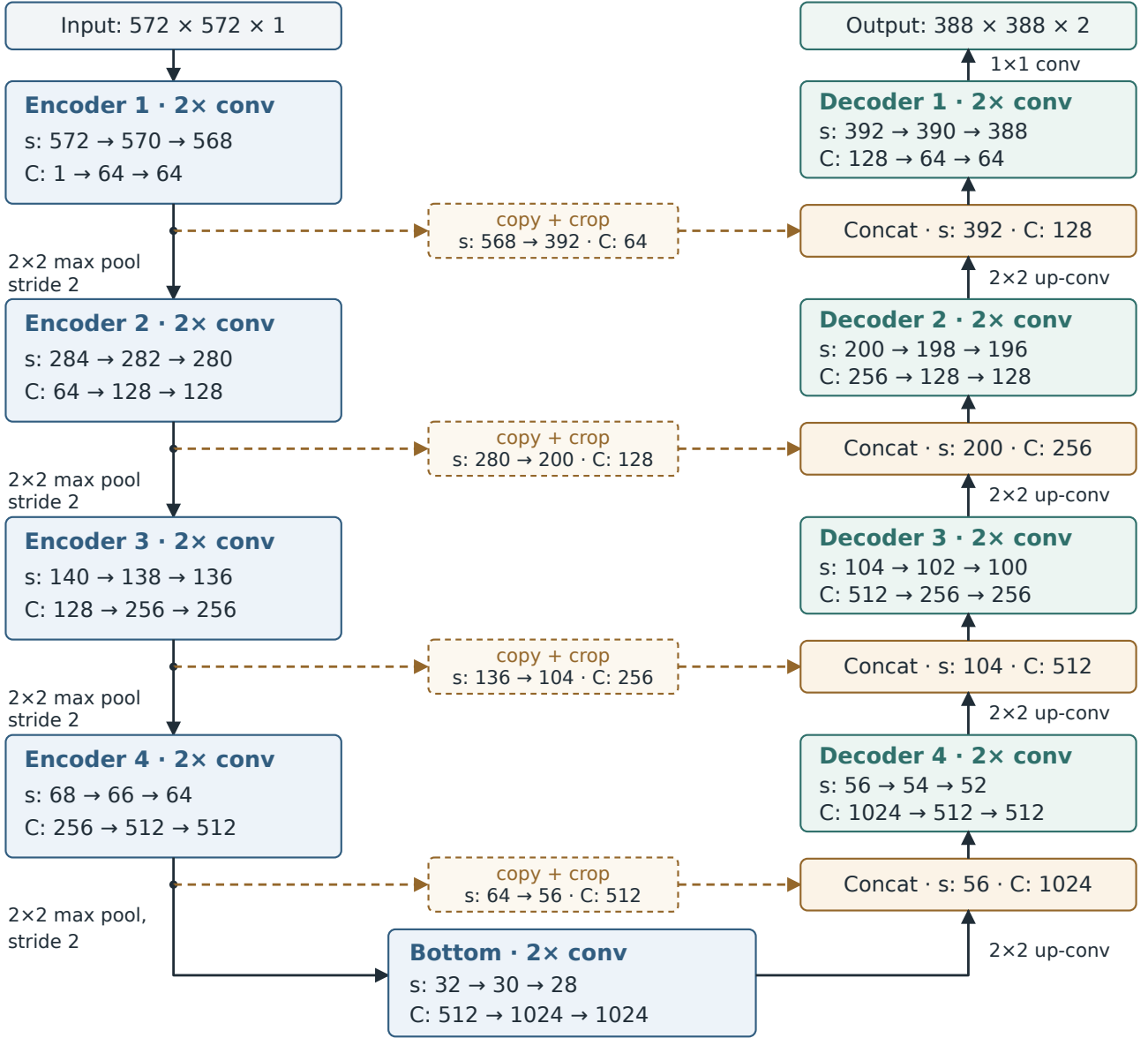
Same spatial resolution; identity shortcuts.
 BN after every conv is omitted for clarity.
 Final BN → addition → ReLU (post-activation).

Figure 2: Original same-shape residual blocks. (a) Basic and (b) bottleneck branches merge elementwise with their own unchanged inputs, then apply ReLU. Batch normalization follows every convolution but is omitted visually; no activation intervenes between final normalization and addition. Neither branch depicts a projection shortcut. Based on He et al. [2].

Original U-Net

2D segmentation · 2015

s = spatial side length (pixels); C = channels. All feature maps are square.



2× conv blocks: two valid 3×3 convolutions, each followed by ReLU.

Up-conv doubles s and halves C . Concat joins channels; it is not addition.

Dashed arrows copy encoder features; cropping aligns their spatial sizes.

Schematic: block sizes do not encode tensor dimensions.

Figure 3: Original valid-convolution U-Net. Dashed paths copy and crop encoder outputs to match the corresponding upsampled maps. Concatenation joins channels rather than adding features. Spatial side lengths s and channel counts C distinguish resolution changes from feature fusion. The illustrated output has two class channels. Based on Ronneberger et al. [3].